TASK01: NETWORKING FUNDAMENTALS, NMAP SCANNING

AND AUTOMATION SCRIPTING

INTRODUCTION:

This task focuses on understanding fundamental networking concepts such as IP addresses, ports, and communication protocols. The objective of this task is to gain hands-on experience in network scanning using Nmap and analyze scan results to identify open ports, running services, and potential security risks.

A controlled lab environment was created using Kali Linux as the attacker machine and Metasploitable as the target machine. Both machines were connected through an isolated internal network in VirtualBox to ensure safe testing.

In addition, a Python automation script was developed using the python-nmap library to automate scanning and generate structured reports. This task helps in building foundational skills required for SOC analysts and cybersecurity professionals.

Networking Fundamentals

IP Address:

An IP address is a unique identifier assigned to devices in a network. It enables communication between systems.

There are two main types:

- IPv4 – 32-bit address (Example: 192.168.100.10)
- IPv6 – 128-bit address

In this lab:

- Kali Linux: 192.168.100.10
- Metasploitable: 192.168.100.20

Ports

Ports allow multiple services to run on a single IP address. Each port is associated with a specific service.

Examples:

- Port 21 → FTP
- Port 22 → SSH
- Port 23 → Telnet
- Port 80 → HTTP

Port states:

- Open → Service is running
- Closed → No service running
- Filtered → Blocked by firewall

TCP vs UDP

TCP (Transmission Control Protocol):

- Connection-oriented
- Reliable
- Uses three-way handshake

UDP (User Datagram Protocol):

- Connectionless
- Faster

- No reliability guarantee

Lab Setup

The lab environment was created using Oracle VirtualBox.

- Kali Linux configured as attacker machine.
- Metasploitable configured as vulnerable target.
- Both machines connected using Internal Network mode.
- Static IP addresses assigned manually.

Kali configuration:

```
sudo ip addr add 192.168.100.10/24 dev eth0
sudo ip link set eth0 up
```

Metasploitable configuration:

```
sudo ifconfig eth0 192.168.100.20 netmask 255.255.255.0 up
```

Connectivity was verified using ping command.

Metasploit [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
msfadmin@metasploitable:~$ sudo ifconfig eth0 192.168.100.20 netmask 255.255.255
.0 up
[sudo] password for msfadmin:
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:09:6b:95
          inet addr:192.168.100.20  Bcast:192.168.100.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe09:6b95/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:49 errors:0 dropped:0 overruns:0 frame:0
          TX packets:28 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:13022 (12.7 KB)  TX bytes:4456 (4.3 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:101 errors:0 dropped:0 overruns:0 frame:0
          TX packets:101 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:23573 (23.0 KB)  TX bytes:23573 (23.0 KB)

msfadmin@metasploitable:~$ _
```

Caps Lock

1 2 3 4 3:23

kali@kali: ~

Session Actions Edit View Help

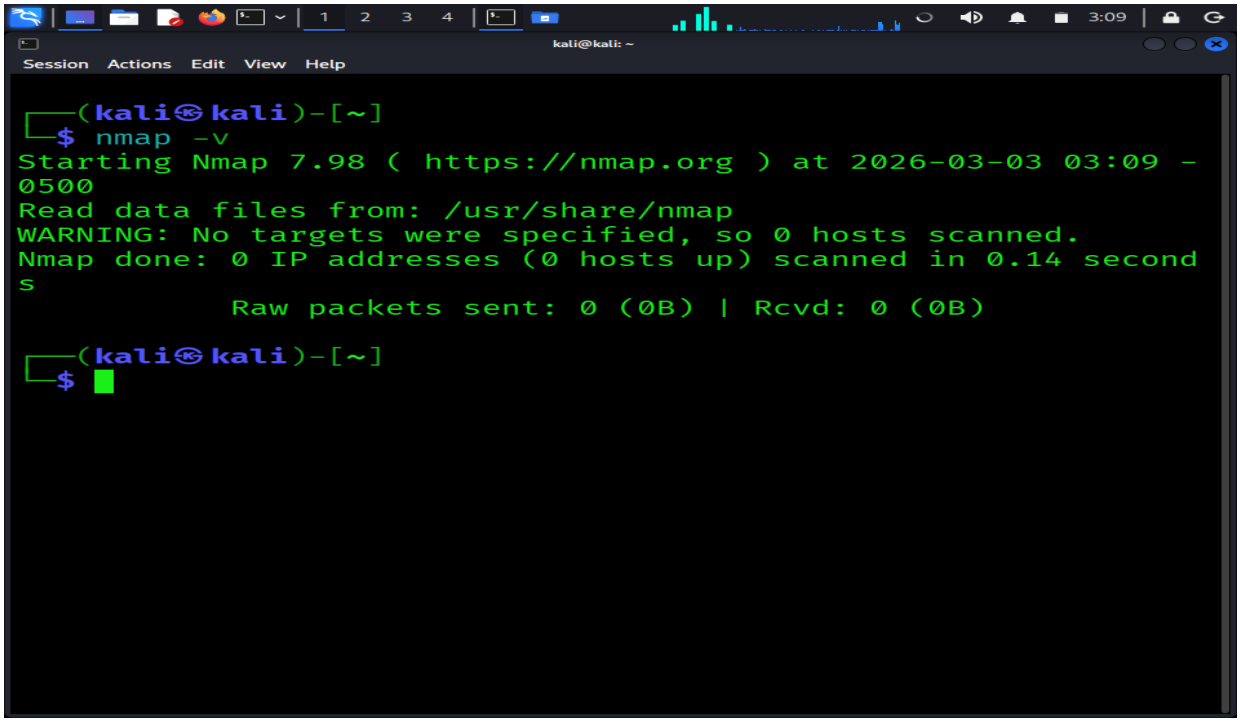
```
┌──(kali㉿kali)-[~]
└─$ sudo ip addr add 192.168.100.10/24 dev eth0

┌──(kali㉿kali)-[~]
└─$ sudo ip link set eth0 up

┌──(kali㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state U
NKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_
codel state UP group default qlen 1000
    link/ether 08:00:27:1f:b7:23 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.10/24 scope global eth0
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdis
c noqueue state DOWN group default
    link/ether 02:42:cf:38:b4:05 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker
0
```

Nmap Installation

Nmap was verified using:

nmap -v

This confirmed that Nmap version 7.98 was installed and ready for use.



Scan Results

SYN Scan

Command used:

sudo nmap -sS 192.168.100.20 -oN syn_scan.txt

This scan performs a half-open scan and is considered stealthy.

Result:
Multiple ports were identified as open including FTP, SSH, Telnet, HTTP, MySQL, and others.

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS 192.168.100.20 -oN syn_scan.txt
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-03 03:25 -
0500
Nmap scan report for 192.168.100.20
Host is up (0.0058s latency).
Not shown: 978 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8180/tcp open  unknown
MAC Address: 08:00:27:09:6B:95 (Oracle VirtualBox virtual NIC
)

Nmap done: 1 IP address (1 host up) scanned in 0.62 seconds

┌──(kali㉿kali)-[~]
└─$
```

```
┌──(kali㉿kali)-[~]
└─$ cat syn_scan.txt
# Nmap 7.98 scan initiated Tue Mar  3 03:25:32 2026 as: /usr/lib/nmap/nmap -sS -oN syn_scan.
txt 192.168.100.20
Nmap scan report for 192.168.100.20
Host is up (0.0058s latency).
Not shown: 978 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8180/tcp open  unknown
MAC Address: 08:00:27:09:6B:95 (Oracle VirtualBox virtual NIC)

# Nmap done at Tue Mar  3 03:25:32 2026 -- 1 IP address (1 host up) scanned in 0.62 seconds

┌──(kali㉿kali)-[~]
└─$
```

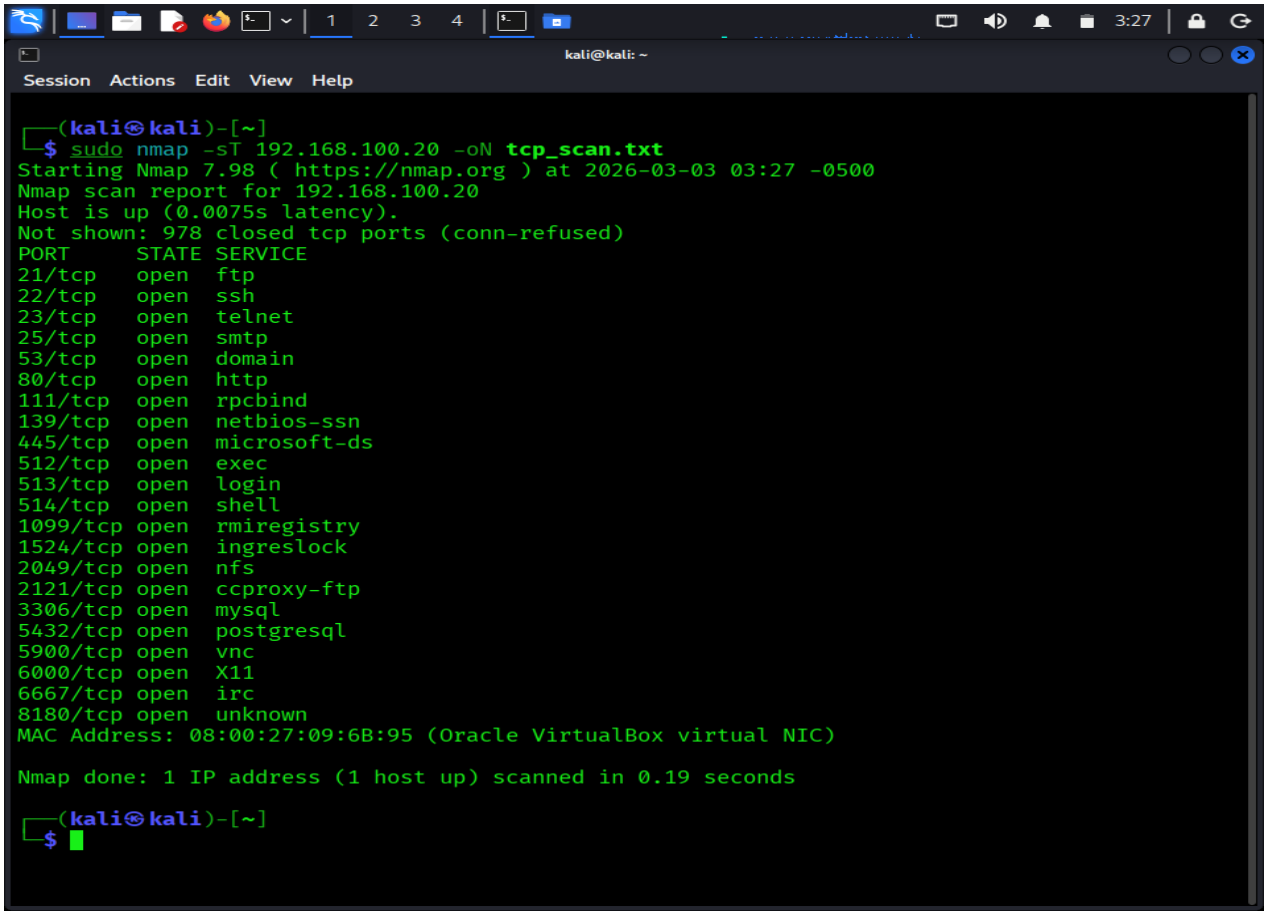
TCP Connect Scan

Command:

sudo nmap -sT 192.168.100.20 -oN tcp_scan.txt

This scan completes the full three-way handshake and is more detectable.

The results were consistent with the SYN scan



UDP Scan

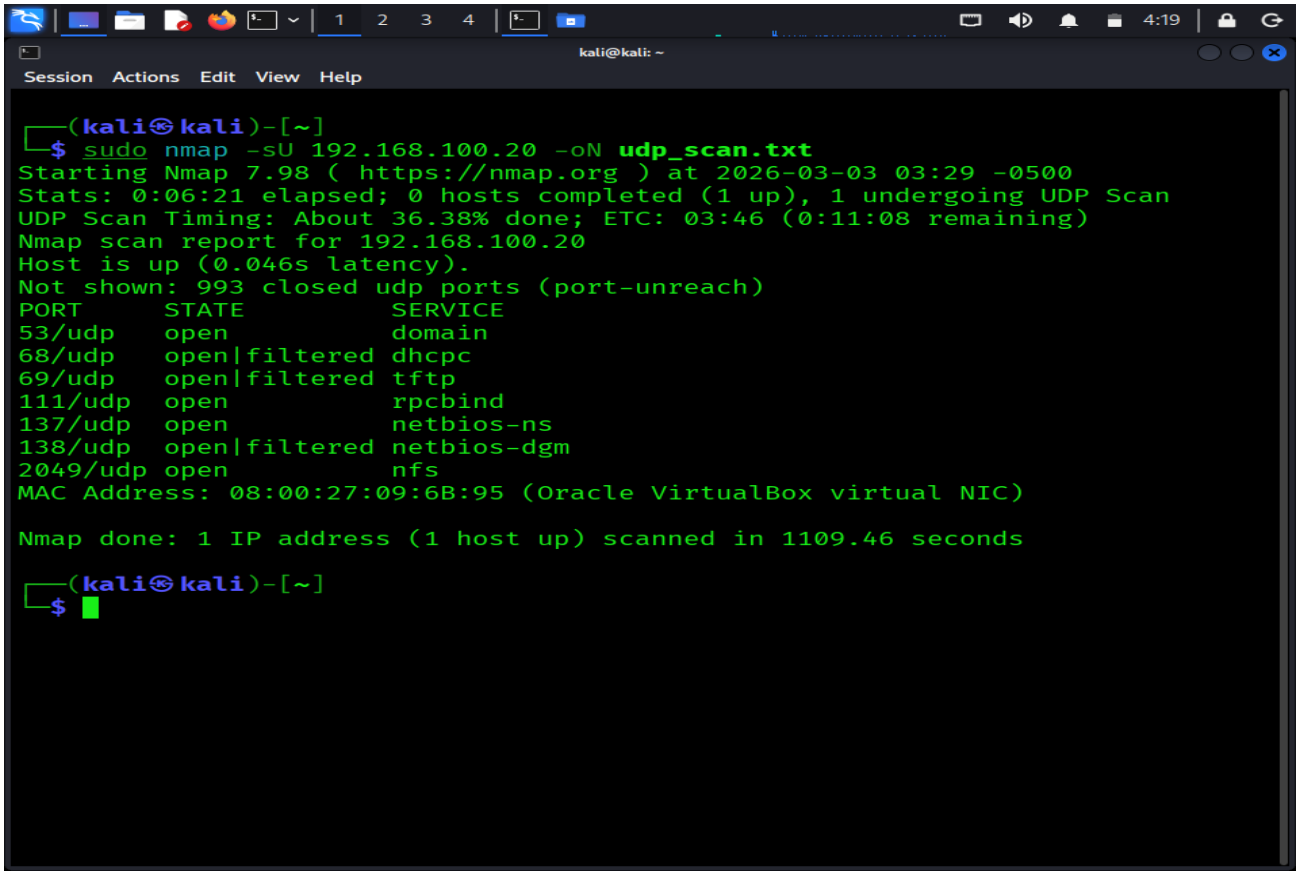
Command:

sudo nmap -sU 192.168.100.20 -oN udp_scan.txt

This scan identifies UDP-based services.

Detected:

- DNS (53)
- NetBIOS
- NFS



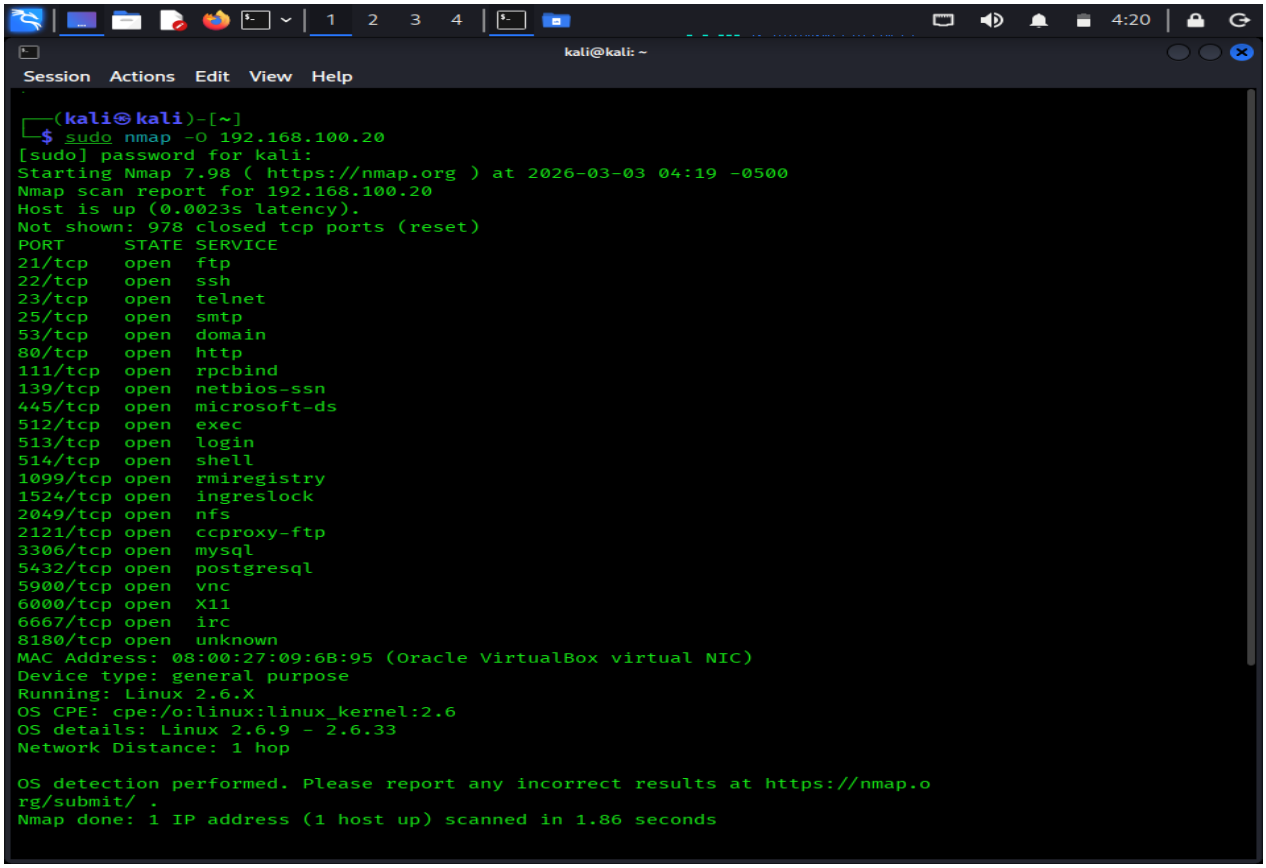
OS Detection

Command:

sudo nmap -O 192.168.100.20

The target was identified as:

Linux Kernel 2.6.9



Service Research

Service 1: FTP

Port: 21
Version: vsftpd 2.3.4

This version contains a known backdoor vulnerability (CVE-2011-2523).

Impact:
An attacker can gain unauthorized remote access, potentially leading to full system compromise.

Service 2: SSH

Port: 22
Version: OpenSSH 4.7p1

Vulnerability: CVE-2008-5161

Impact:
Potential leakage of encrypted data due to CBC mode weakness

Risk Analysis

Based on the scan results, the following risks were identified:

Port	Service	Risk Level	Reason
21	FTP	Critical	Backdoor vulnerability
22	SSH	Medium	CBC encryption weakness
23	Telnet	High	Unencrypted communication
3306	MySQL	High	Database exposure risk
5432	PostgreSQL	High	Database exposure

Recommendations:

- Disable unused services
- Update outdated software
- Use firewall restrictions
- Implement strong authentication
- Conduct regular vulnerability scans

Python Automation

A Python script was developed using python-nmap to automate scanning.

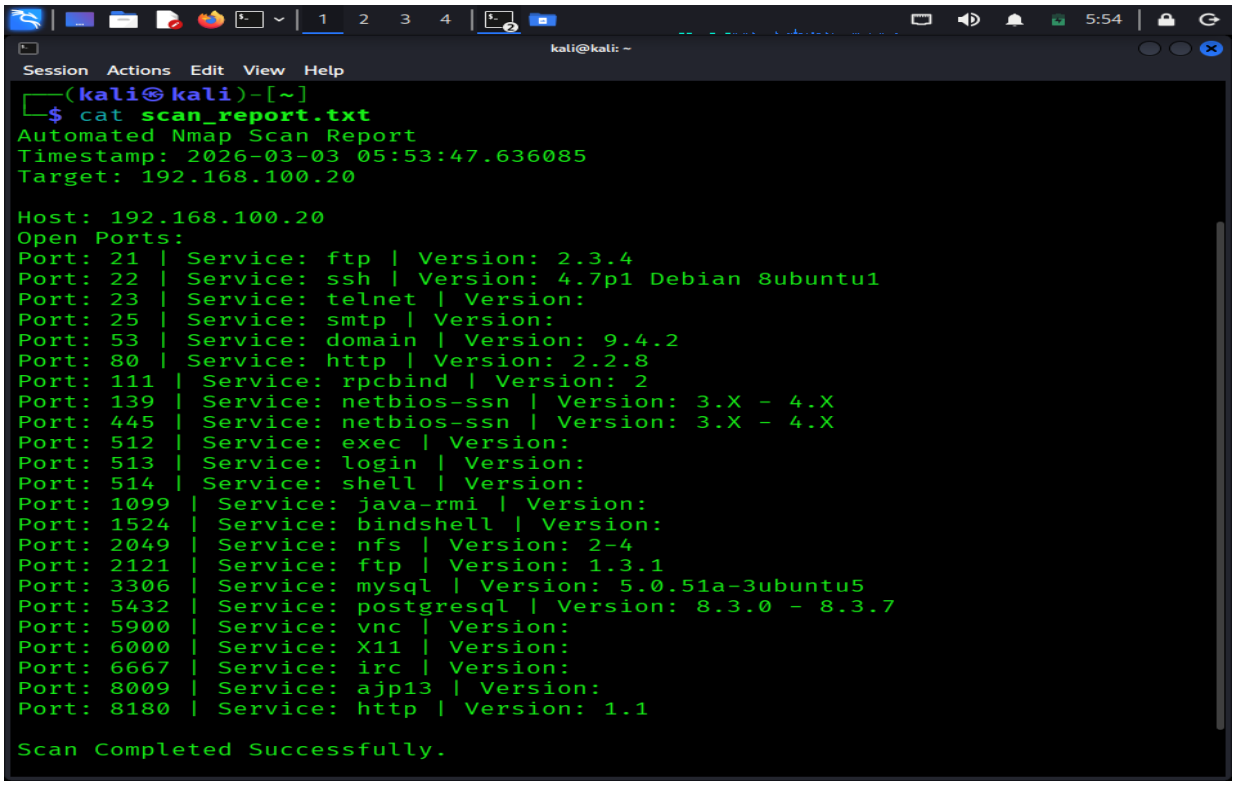
The script:

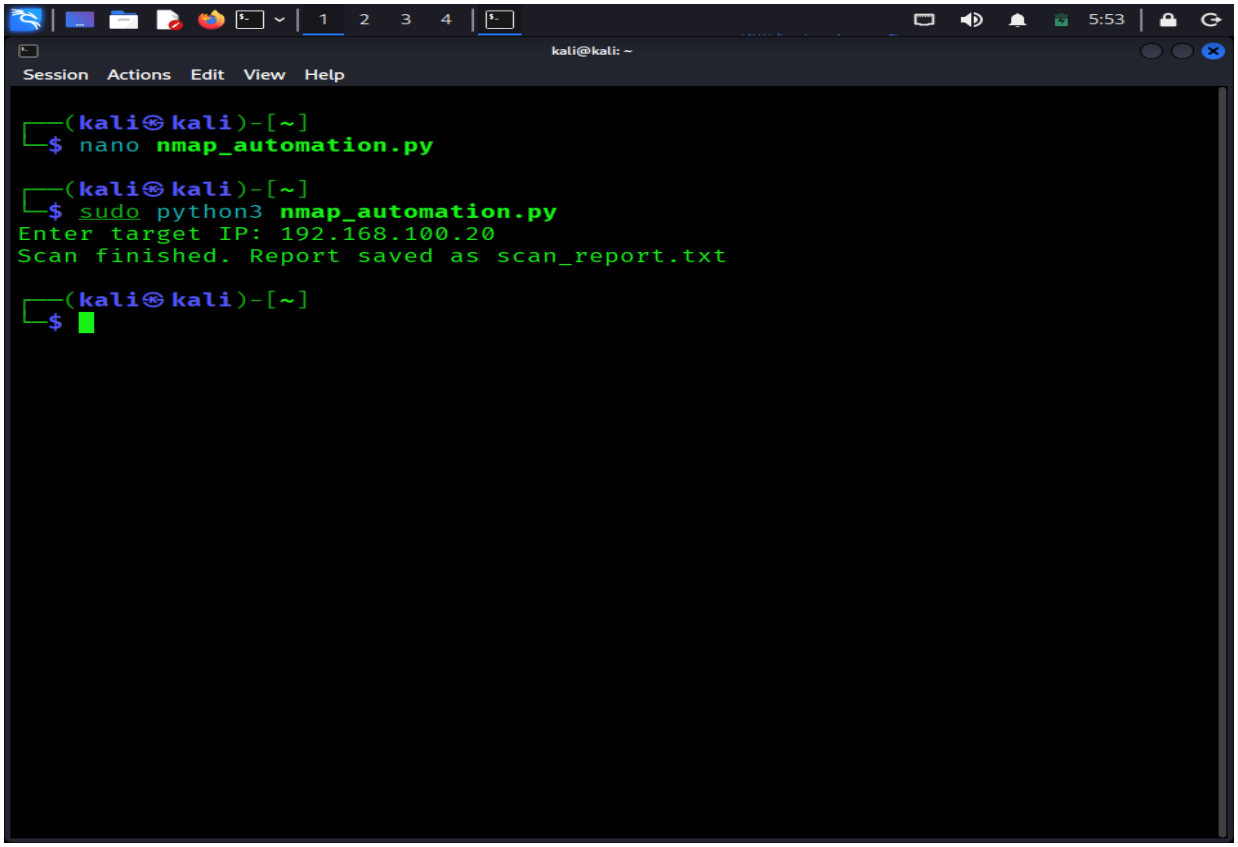
- Accepts target IP as input
- Performs SYN and version scan
- Extracts open ports
- Generates a text report
- Includes timestamp

Execution:

sudo python3 nmap_automation.py

The script successfully generated scan_report.txt containing structured scan results.

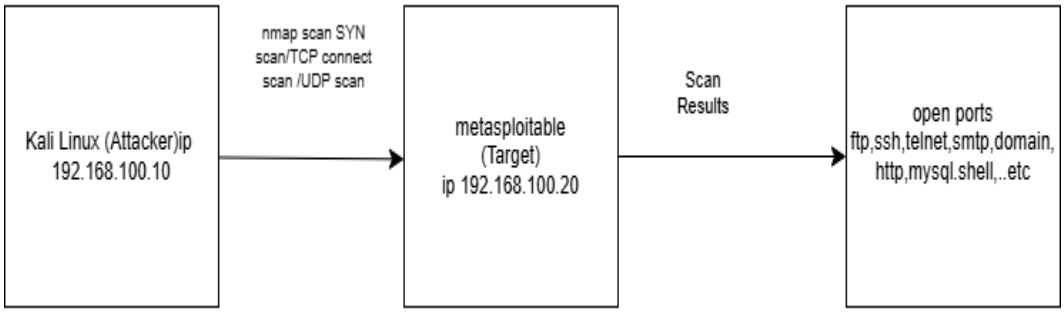




Network Diagram

The network diagram shows:

- Kali Linux (Attacker) – 192.168.100.10
- Metasploitable (Target) – 192.168.100.20
- Nmap scanning direction
- Open services identified



Conclusion

This task provided practical experience in network scanning and vulnerability identification. By setting up an isolated lab environment, performing multiple types of Nmap scans, analyzing results, researching vulnerabilities, and developing a Python automation script, essential cybersecurity skills were developed.

This exercise strengthened understanding of:

- Networking fundamentals
- Port and service identification
- Risk assessment
- Vulnerability research
- Automation scripting

These skills are fundamental for SOC analysts and penetration testers in real-world environments.